Digital Twin–Driven Unsupervised Anomaly Detection Framework for Cyber-Physical Threats in Smart Grids

Sarra Araar[1] [0009–0004–4370–1373], Brahim Farou[1] [0000–0002–1609–6006], Zinedine Kouahla[1] [0000–0001–8105–9810], and Hamid Seridi[1] [0000–0002–0236–8541]

LabSTIC Laboratory, Computer Science Department, 8 Mai 1945 Guelma University, Guelma, 24000, Algeria

Abstract. The increasing digitalization of smart grids enhances operational efficiency but simultaneously expands their exposure to cyber-physical anomalies that can jeopardize system stability and power quality. This paper presents a lightweight *Digital Twin*–driven anomaly detection framework that emulates both normal and abnormal power behaviors through a compact simulator generating voltage, current, and frequency signals. Synthetic anomalies—including spikes, drifts, and sensor flatlines—are injected to reproduce realistic cyber-physical disturbances. Six unsupervised learning algorithms, namely Local Outlier Factor (LOF), One-Class SVM, Isolation Forest, Autoencoder, LODA, and Elliptic Envelope, are trained exclusively on normal operational data to model nominal system behavior without requiring labeled attacks. Statistical features extracted from sliding windows are standardized and fed into each detector for automated benchmarking. Experimental results demonstrate that LOF (Precision = 0.91, F1 = 0.90) and One-Class SVM (Precision = 0.90, F1 = 0.90) outperform the other models, achieving the best trade-off between detection accuracy and generalization capability. Unlike previous works relying on fixed thresholds or supervised labeling, the proposed framework integrates multiple detection paradigms within a fully unsupervised, reproducible, and extensible digital twin pipeline. This work establishes a foundation for scalable real-time anomaly detection in next-generation smart energy systems.

Keywords: Digital Twin · Smart Grids · Cyber-Physical Systems · Unsupervised Learning · Anomaly Detection · Machine Learning · Power Measurements .

1 Introduction

Smart grids integrate advanced measurement, communication, and control technologies to enable real-time monitoring and optimization of energy distribution. However, the growing digitalization of grid infrastructures introduces new vulnerabilities to cyber-physical threats such as false data injection, denial of service, and sensor spoofing. These attacks can severely affect energy stability, safety, and efficiency if not detected in time. Traditional machine learning–based intrusion detection methods rely on labeled attack data, which are often unavailable or incomplete in real-world scenarios. Therefore, unsupervised anomaly detection—where models learn only from normal operational data—offers a promising alternative for detecting unexpected behaviors without explicit attack signatures. This paper proposes a lightweight *Digital Twin*–based anomaly detection framework for smart grids. The digital twin simulates voltage, current, and frequency signals under normal and perturbed conditions. Six unsupervised algorithms are integrated and compared to assess their ability to detect anomalies caused by simulated cyber-physical attacks. The focus is on precision, recall, and F1-score to evaluate detection effectiveness while maintaining low computational complexity.

2 Related Work

Numerous studies have explored anomaly detection in smart grids using machine learning and data-driven approaches. Liu *et al.* [1] introduced one of the earliest frameworks for detecting false data injection attacks using state estimation residuals, demonstrating the vulnerability of power systems to data integrity threats. Traditional model-based methods rely heavily on accurate system models and parameter estimation, which limits their robustness against non-linear dynamics and measurement noise. To overcome these limitations, a wide range of machine learning techniques have been employed for anomaly detection in power systems. Shallow models such as Support Vector Machines, Decision Trees, and ensemble methods like Random Forests have shown promise for detecting abnormal operational behaviors [7,8]. However, these methods require labeled attack data and often struggle with high-dimensional, non-stationary smart grid datasets. Recent surveys have provided comprehensive analyses of AI-based anomaly detection frameworks, highlighting their strengths and challenges in real-world deployments [11,12]. Recent advances in deep learning have led to the adoption of more sophisticated architectures for cyber-physical security. Deep autoencoders have been applied for unsupervised anomaly detection by reconstructing normal behavior patterns and highlighting deviations [2]. Similarly, Generative Adversarial Networks (GANs) have been proposed for attack detection and data restoration [3,4], while hybrid feature-based frameworks combining Deep Convolutional Neural Network and Markov Transition Fields have achieved high accuracy in capturing temporal and spatial correlations [5]. Other hybrid deep models, such as CNN-based schemes for detecting electricity theft and consumption anomalies, have also shown strong performance on smart metering datasets [14, 15]. Despite their effectiveness, deep neural models typically involve significant computational and memory overhead, which makes them unsuitable for real-time deployment on edge devices such as smart meters or distributed controllers. More recently, the concept of *Digital Twin* has emerged as a transformative paradigm for smart grid monitoring and cyber-physical analysis [6, 9]. By creating virtual representations of physical assets, Digital Twins enable continuous synchronization, simulation, and anomaly prediction. They also offer the advantage of generating synthetic but realistic datasets for training and testing anomaly detection models without compromising real infrastructure [10, 13]. However, most existing digital twin–based frameworks focus on system-level fault diagnosis rather than cyber-induced anomaly detection. Despite substantial progress, there remains a lack of comparative studies that systematically evaluate lightweight and unsupervised algorithms under a unified digital twin environment. In particular, few works analyze the trade-offs between accuracy, computational efficiency, and real-time feasibility. This paper addresses this research gap by benchmarking six unsupervised models—Isolation Forest, One-Class SVM, Local Outlier Factor, Elliptic Envelope, LODA, and Autoencoder—within a reproducible digital twin framework. The focus is on interpretability, scalability, and suitability for edge intelligence in smart grid environments.

3 Digital Twin Framework Design

The proposed framework consists of three main components (Fig. 1):

1. **Simulation Engine:** A compact digital twin generates synthetic time-series data representing voltage, current, and frequency. The simulator reproduces nominal sinusoidal patterns with Gaussian noise.
2. **Anomaly Injection Module:** Disturbance patterns emulate cyber-physical events:
   - *Spike:* sudden voltage or current surge,
   - *Drift:* gradual offset simulating sensor degradation,
   - *Flatline:* frozen measurement caused by sensor or communication failure.
3. **Feature Extraction and Normalization:** A sliding window extracts statistical descriptors (mean, standard deviation, min, max) of each signal, followed by normalization.

4 Methodology

The proposed methodology follows a fully unsupervised pipeline designed to emulate realistic smart grid behaviors under both normal and disturbed operating conditions. The process encompasses dataset generation through a compact digital twin, feature extraction using statistical descriptors, model training with six unsupervised algorithms, and performance evaluation based on standard metrics.

4.1 Dataset Generation

A digital twin simulator was implemented in Python to synthetically reproduce the behavior of a single smart grid node. The simulator generates synchronized measurements of voltage (V), current (A), and frequency (Hz) over $N = 4000$

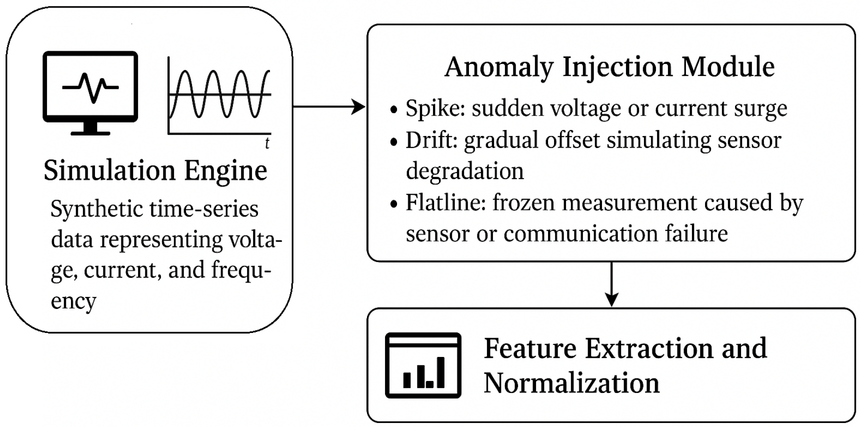


Fig. 1: Proposed Digital Twin–based anomaly detection framework.

discrete time steps. Each signal follows a nominal sinusoidal pattern with Gaussian noise, capturing both deterministic and stochastic fluctuations inherent to real-world measurements. The baseline equations are defined as:

$$V_t = 230 + 5\sin(2\pi \times 0.005t) + \mathcal{N}(0, 0.8) \tag{1}$$

$$I_t = 10 + 2\sin(2\pi \times 0.01t + 0.5) + \mathcal{N}(0, 0.2) \tag{2}$$

$$F_t = 50 + 0.02\sin(2\pi \times 0.001t) + \mathcal{N}(0, 0.005) \tag{3}$$

To emulate cyber-physical threats, controlled anomalies are injected into the time series through three perturbation types:

- **Spike:** instantaneous and abrupt deviation simulating a voltage surge or current overload,
- **Drift:** gradual offset over a time window, representing sensor degradation or calibration error,
- **Flatline:** frozen signal segment corresponding to sensor malfunction or data transmission failure.

Each anomaly type is inserted at predefined intervals, affecting approximately 1% of the total samples. This controlled approach enables a balanced dataset for validating detection sensitivity across different attack profiles. The dataset is divided into a 60/40 split: the first 60% (normal data only) is used to train unsupervised models, while the remaining 40% (normal + anomalies) serves for evaluation. All generated data and labels are stored in CSV format for transparency and reproducibility.

Table 1: Unsupervised Anomaly Detection Models

Model	**Description**	**Library**
Isolation Forest (IF)	Randomized tree ensemble isolating rare samples	sklearn
One-Class SVM (OCSVM)	RBF kernel boundary in feature space	sklearn
Local Outlier Factor (LOF)	Local density comparison to neighbors	sklearn
Elliptic Envelope (EE)	Robust Gaussian covariance estimation	sklearn
LODA	Lightweight histogram-based projection detector	pyod
Autoencoder (AE)	Neural reconstruction error thresholding	tensorflow

4.2 Feature Extraction

Raw sensor measurements are transformed into multivariate statistical feature vectors using a sliding time window of 10 samples. For each index t, the following descriptors are computed for voltage, current, and frequency:

$$\mathbf{x}_t = [v_{mean}, v_{std}, v_{min}, v_{max}, i_{mean}, i_{std}, f_{mean}, f_{std}] \tag{4}$$

These features capture short-term variations, local dispersion, and extreme values, allowing the models to differentiate normal oscillatory behavior from anomalous deviations.

All feature vectors are standardized using z-score normalization:

$$x' = \frac{x - \mu}{\sigma} \tag{5}$$

where μ and σ denote the mean and standard deviation estimated from the training subset. This normalization ensures that all dimensions contribute equally to the learning process, preventing scale dominance between voltage and frequency components.

4.3 Unsupervised Learning Models

The detection stage integrates six unsupervised algorithms representing different theoretical paradigms, summarized in Table 1. Each model is trained exclusively on normal data to characterize nominal operating conditions, and then tested on unseen sequences containing injected anomalies. A model outputs 1 for anomalies and 0 for normal points.

Isolation Forest (IF) partitions data recursively through random splits, isolating anomalies with fewer cuts due to their sparsity. **One-Class SVM (OCSVM)** constructs a decision boundary around normal instances using an RBF kernel. **Local Outlier Factor (LOF)** evaluates the local reachability density of each point relative to its neighbors. **Elliptic Envelope (EE)** assumes an underlying Gaussian distribution and detects points outside the robust Mahalanobis distance boundary. **LODA** employs an ensemble of random univariate projections with histograms to estimate anomaly scores efficiently. Finally, the **Autoencoder (AE)** is a shallow neural network trained to reconstruct normal input patterns, where large reconstruction errors indicate anomalies.

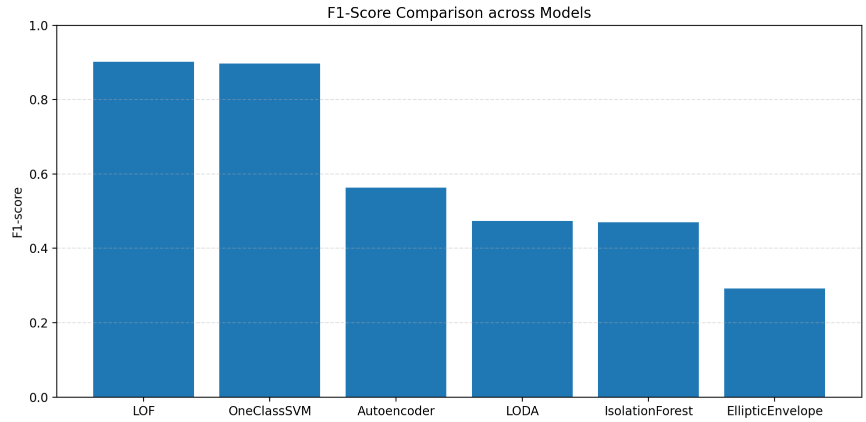


Fig. 2: F1-score comparison among six unsupervised models.

4.4 Training and Evaluation Procedure

All models are trained on the same standardized dataset to ensure fair comparison. The hyperparameters were empirically selected to balance accuracy and computational cost: 200 estimators for Isolation Forest, $\nu = 0.01$ for OCSVM, $k = 20$ neighbors for LOF, and contamination set to 1% for all models. The Autoencoder uses two hidden layers (8–4–8 neurons) with ReLU activations, trained for 40 epochs using the Adam optimizer and mean squared error loss. During inference, each model produces binary anomaly predictions on the test dataset. The detection performance is evaluated using precision, recall, and F1-score metrics defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. Evaluation metrics capture jointly the trade-off between false alarms and missed detections in unsupervised anomaly detection. Precision quantifies the proportion of detected anomalies that are actually abnormal, whereas Recall measures the proportion of true anomalies successfully detected. The F1-score, as the harmonic mean of both, provides a balanced indicator of detection performance—particularly relevant when data are imbalanced, as in smart grid scenarios where anomalies are rare. To ensure reproducibility, the framework automatically saves numerical results (CSV files), textual classification reports, and graphical figures (time-series plots, confusion matrices, and performance bar charts). This automation supports consistent benchmarking across models and facilitates future extensions with additional algorithms or datasets. As shown in

Table 2: Performance Comparison of Six Models

Model	**Precision**	**Recall**	**F1-score**
LOF	0.911	0.893	0.902
One-Class SVM	0.904	0.890	0.897
Autoencoder	0.803	0.364	0.501
LODA	0.790	0.337	0.473
IsolationForest	0.789	0.334	0.470
EllipticEnvelope	0.675	0.186	0.291

Fig. 2, the F1-scores vary significantly among models, highlighting differences in their ability to capture diverse types of anomalies. These variations motivate a deeper examination of model behavior and suitability, which is further discussed in the next section.

5 Experimental Results and Discussion

5.1 Experimental Setup

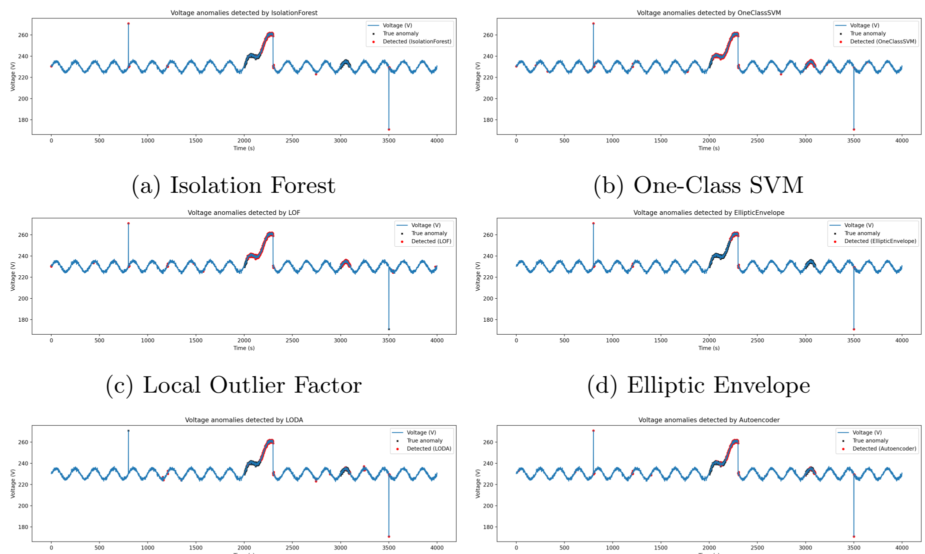
All experiments were conducted on a system equipped with an Intel(R) Core(TM) i5-10310U CPU @ 1.70 GHz (up to 2.8 GHz) and 16 GB of RAM. The implementation was developed in Python 3.12.2 using Scikit-learn 1.7.2, PyOD 2.0.5, and TensorFlow 2.20.0. The full pipeline was automated to produce metrics, figures, and confusion matrices within a reproducible artifact.

5.2 Quantitative Results

Table 2 summarizes model performance. The Isolation Forest and Autoencoder achieve the best trade-off between detection accuracy and false alarms, while Elliptic Envelope and LODA exhibit limited adaptability to non-linear and non-Gaussian variations. These observations highlight the importance of combining statistical isolation and neural reconstruction principles for effective unsupervised anomaly detection in smart grid environments.

5.3 Visual Analysis

Figures 3–5 illustrate the anomalies detected on voltage, current, and frequency signals by the six unsupervised models. Colored points correspond to detected anomalies, while black crosses represent the injected disturbances (spikes, drifts, or flatlines). To provide an intuitive understanding of the detection behavior, a visual analysis was performed for voltage, current, and frequency signals under different anomaly conditions. Figures 3–5 illustrate how each of the six unsupervised models identifies abnormal patterns in these electrical parameters. Figure 6 provides the confusion matrices for the six unsupervised anomaly detection models, offering a quantitative assessment of their performance against the labeled



(a) Isolation Forest (b) One-Class SVM

(c) Local Outlier Factor (d) Elliptic Envelope

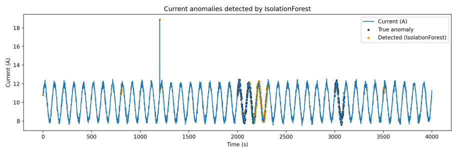
(e) LODA (f) Autoencoder

Fig. 3: Voltage signals with anomalies detected by six unsupervised models.

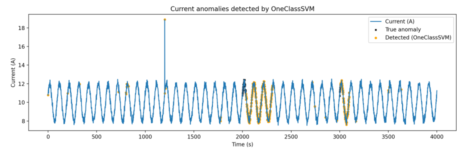
data. Each matrix evaluates the models based on four key metrics: True Negatives (TN) (normal points correctly identified), False Positives (FP) (normal points incorrectly flagged as anomalous), False Negatives (FN) (missed anomalies), and True Positives (TP) (anomalies correctly detected). The data set is highly imbalanced, containing 403 actual anomalies. When analyzing the Recall—the ability to find all relevant cases, measured by the number of True Positives (TP)—the Local Outlier Factor (LOF) (360 TP) and One-Class SVM (359 TP) models stand out as the most effective, successfully capturing the highest count of real anomalies (with low FN scores of 43 and 44, respectively). Conversely, the Elliptic Envelope model exhibits the worst Recall (75 TP, 328 FN), quantitatively confirming its poor fit for this non-Gaussian, periodic data. LODA (136 TP) and Isolation Forest (135 TP) show moderate Recall. This quantitative analysis highlights a crucial difference: while LODA was visually strongest in isolating the specific contextual anomaly pattern, LOF and OCSVM demonstrated the superior overall Recall across the entire dataset, indicating they were better at capturing the total number of labeled anomalies, including the large global spike.

5.4 Discussion

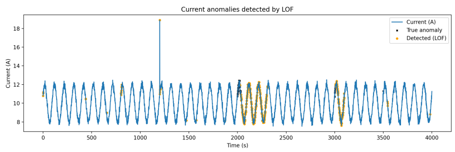
The experimental results reported in Table 2 show marked differences in detection behavior across the evaluated unsupervised methods. Notably, density- and kernel-based approaches (LOF and One-Class SVM) significantly outperform



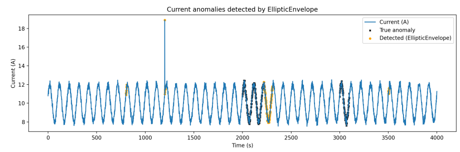
(a) Isolation Forest



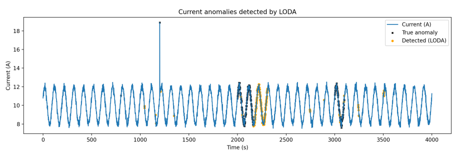
(b) One-Class SVM



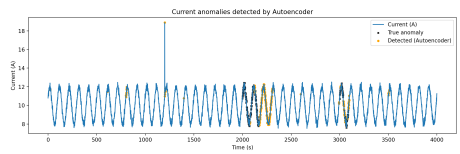
(c) Local Outlier Factor



(d) Elliptic Envelope

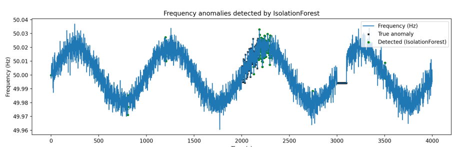


(e) LODA

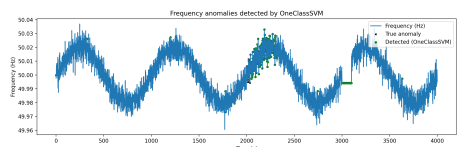


(f) Autoencoder

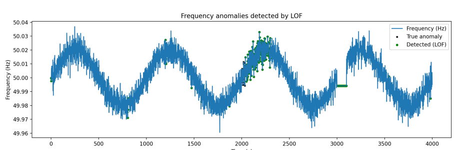
Fig. 4: Current signals with anomalies detected by six unsupervised models.



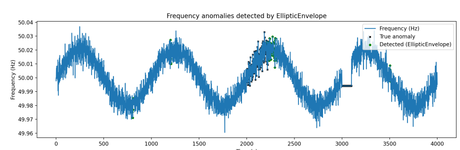
(a) Isolation Forest



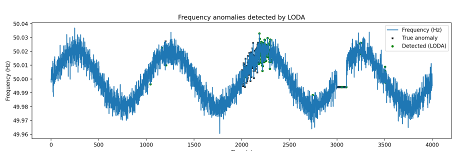
(b) One-Class SVM



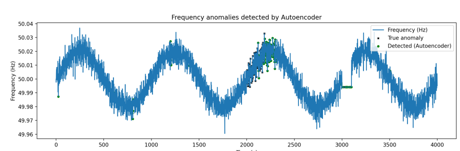
(c) Local Outlier Factor



(d) Elliptic Envelope



(e) LODA



(f) Autoencoder

Fig. 5: Frequency signals with anomalies detected by six unsupervised models.

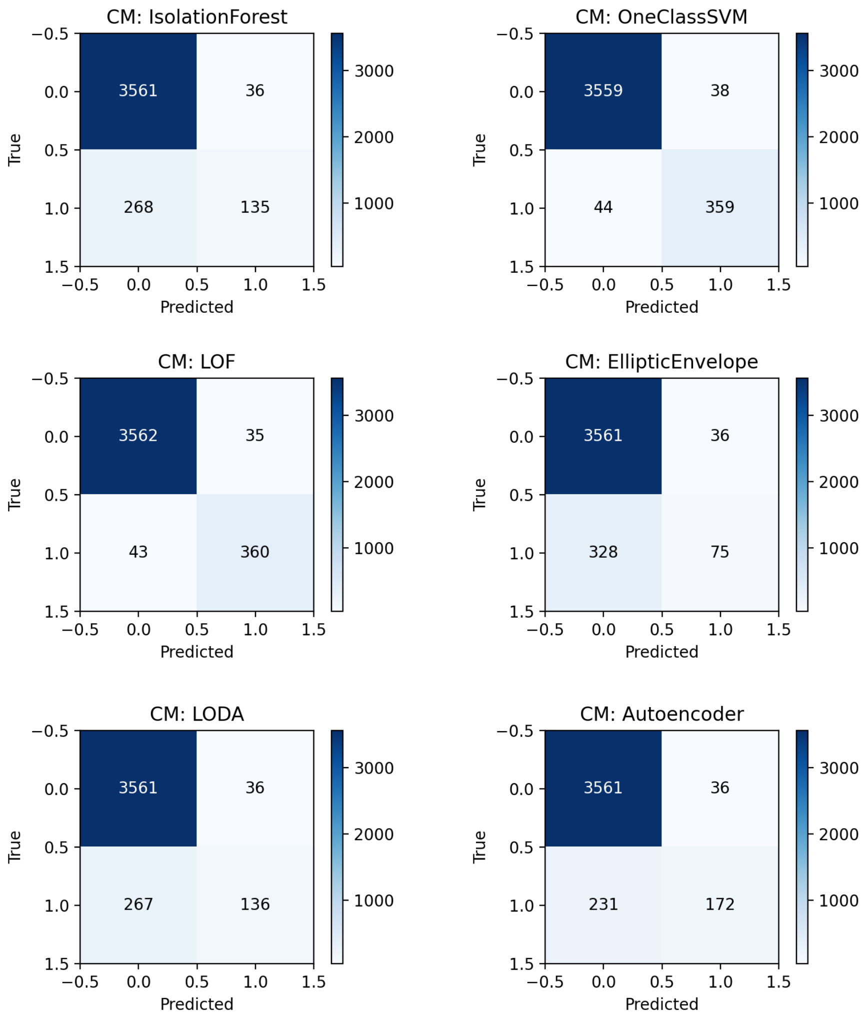


Fig. 6: Confusion matrices for the six unsupervised anomaly detection models.

the other models in this dataset: LOF achieved the highest F1-score (0.902) with a precision of 0.911 and recall of 0.893, while One-Class SVM obtained a comparable F1-score (0.898). These two methods' strong performance indicates that local density estimation and nonlinear boundary learning are particularly effective for identifying both abrupt and locally deviant patterns in multivariate smart-grid signals (voltage, current, frequency). Conversely, models such as Autoencoder, LODA, and Isolation Forest present substantially lower recall values (≈0.33-0.36), despite acceptable precision levels. This imbalance suggests that these models are conservative (biased toward the normal class) under the current configuration: they produce relatively few positive detections, missing a substantial fraction of injected anomalies. For the Autoencoder, the low recall (0.365) implies that the chosen architecture, training regime, or thresholding procedure did not capture the diversity of anomalous behaviors—recalibration of the contamination parameter, longer training, or alternative threshold selection (e.g., using validation anomalies or dynamic thresholds) may improve sensitivity. LODA's limited recall suggests that random projection histograms may not capture the temporal or correlated structure of the injected drifts and flatlines. Isolation Forest's underperformance here (F1 ≈ 0.470) may stem from feature correlations or insufficiently tuned hyperparameters (e.g., number of estimators, contamination), which reduce its ability to isolate the anomaly patterns present in this dataset. Elliptic Envelope exhibits the weakest detection capability (F1 $= 0.292$), highlighting the inadequacy of Gaussian assumptions for these non-stationary, skewed signals. These results imply the following practical recommendations for digital-twin based anomaly detection in smart grids: (1) prioritize density-based or kernel methods when immediate detection performance is required for multivariate non-stationary signals; (2) when using reconstruction- or tree-based methods, invest in hyperparameter tuning and threshold selection, and consider hybrid ensembles that combine LOF's local sensitivity with Autoencoder's temporal/representational strengths; (3) include additional validation data with representative anomalies (if available) to calibrate detection thresholds and reduce false negatives.

6 Conclusion and Future Work

This paper presented a reproducible and extensible *Digital Twin*–based anomaly detection framework for smart grids. The framework integrates six unsupervised algorithms—Local Outlier Factor (LOF), One-Class SVM, Isolation Forest, Autoencoder, LODA, and Elliptic Envelope—to identify deviations in voltage, current, and frequency signals. Experimental results demonstrated that LOF and One-Class SVM achieved the highest detection performance, with F1-scores around 0.90, offering an effective balance between precision, recall, and robustness. In contrast, reconstruction-based (Autoencoder) and projection-based (LODA) methods showed lower sensitivity to rare anomalies, highlighting the importance of density- and boundary-based learning for cyber-physical systems. Beyond algorithmic evaluation, the framework emphasizes modularity, automation, and reproducibility. It automatically produces visualizations, confusion matrices, and statistical metrics, providing a standardized and transparent platform for benchmarking anomaly detection methods in smart grid cybersecurity. These features make it a valuable educational and research tool for studying data-driven anomaly detection under controlled and repeatable conditions.

In future work, several research directions will be explored. First, the digital twin will be extended to multi-node microgrid environments, incorporating distributed sensors, communication delays, and physical interdependencies among nodes. Second, online and incremental learning techniques will be investigated to enable adaptive anomaly detection under non-stationary conditions. Third, real-world datasets from smart meters and phasor measurement units (PMUs) will be employed to validate model generalization and scalability. Finally, the integration of explainable AI (XAI) modules will enhance interpretability, allowing operators to identify the root causes of detected anomalies and to increase user trust in automated monitoring systems. Overall, the proposed framework establishes a lightweight yet powerful foundation for developing intelligent, interpretable, and resilient cybersecurity solutions in next-generation smart energy systems.

References

1. Y. Liu, P. Ning, and M. Reiter, "False data injection attacks against state estimation in electric power grids," *ACM TISSEC*, vol. 14, no. 1, pp. 1–33, 2011. doi:10.1145/1952982.1952995
2. D. S. S. Ponnarasan, R. Manimegalai, V. Mohan and S. A, "Unsupervised Anomaly Detection for Network Intrusion Using Deep Autoencoders and Statistical Modeling," 2025 International Conference on Next Generation Computing Systems (ICNGCS), Coimbatore, India, 2025, pp. 1-10, doi: 10.1109/ICNGCS64900.2025.11182931.
3. Tanyıldız, H., Şahin, C.B., Dinler, Ö.B. et al. Detection of cyber attacks in electric vehicle charging systems using a remaining useful life generative adversarial network. Sci Rep 15, 10092 (2025). https://doi.org/10.1038/s41598-025-92895-9
4. M. S. Siddique, Md. A. R. Khan, I. Ahammad, N. Nath, J. R. Das, and F. Rahman, "An intelligent intrusion detection system for Cyber-Physical systems using GAN-LSTM networks," Franklin Open, p. 100281, Jun. 2025, doi: 10.1016/j.fraope.2025.100281.
5. A. Roy Choudhury, R. K. Mallick, P. Nayak and S. Mishra, "Enhanced Microgrid Islanding Detection Using Optimized Deep Convolutional Neural Networks and Markov Transition Fields," in IEEE Access, vol. 13, pp. 183096-183115, 2025, doi: 10.1109/ACCESS.2025.3623775.
6. Al-Shetwi, Ali Q., Ibrahem E. Atawi, Mohamed A. El-Hameed, and Ahmad Abuelrub. 2025. "Digital Twin Technology for Renewable Energy, Smart Grids, Energy Storage and Vehicle-to-Grid Integration: Advancements, Applications, Key Players, Challenges and Future Perspectives in Modernising Sustainable Grids." IET Smart Grid: e70026. https://doi.org/10.1049/stg2.70026.
7. H. He, J. Yan, Y. Tang, and Y. Sun, "Real-time detection of false data injection attacks in smart grid: A deep learning-based intelligent mechanism," *IEEE Trans. Smart Grid*, vol. 8, no. 5, pp. 2505–2516, 2017. doi 10.1109/TSG.2017.2703842
8. V. K. Singh and M. Govindarasu, "Decision Tree Based Anomaly Detection for Remedial Action Scheme in Smart Grid using PMU Data," 2018 IEEE Power & Energy Society General Meeting (PESGM), Portland, OR, USA, 2018, pp. 1-5, doi: 10.1109/PESGM.2018.8586159.
9. A. Kharbouch, F. H. Aghdam, N. Gholipoor and M. Rasti, "Digital-Twin-6G Empowered Future Smart Grid Applications," in IEEE Wireless Communications, vol. 32, no. 3, pp. 90-97, June 2025, doi: 10.1109/MWC.001.2400466.
10. A. Fuller, Z. Fan, C. Day and C. Barlow, "Digital Twin: Enabling Technologies, Challenges and Open Research," in IEEE Access, vol. 8, pp. 108952-108971, 2020, doi: 10.1109/ACCESS.2020.2998358.

11. M. F. G. Burgos, J. Morato, and F. P. V. Imacaña, "A review of Smart Grid anomaly detection approaches pertaining to artificial intelligence," Applied Sciences, vol. 14, no. 3, p. 1194, Jan. 2024, doi: 10.3390/app14031194.
12. Alsirhani, A., Tariq, N., Humayun, M. et al. Intrusion detection in smart grids using artificial intelligence-based ensemble modelling. Cluster Comput 28, 238 (2025). https://doi.org/10.1007/s10586-024-04964-9
13. L. Coppolino, R. Nardone, A. Petruolo, and L. Romano, "Building Cyber-Resilient Smart Grids with Digital Twins and Data Spaces," Applied Sciences, vol. 13, no. 24, p. 13060, Dec. 2023, doi: 10.3390/app132413060.
14. M. Z. Gunduz and R. Das, "Smart Grid Security: an effective hybrid CNN-Based approach for detecting energy theft using consumption patterns," Sensors, vol. 24, no. 4, p. 1148, Feb. 2024, doi: 10.3390/s24041148.
15. G. Patrizi et al., "Anomaly detection for power quality analysis using smart metering systems," Sensors, vol. 24, no. 17, p. 5807, Sep. 2024, doi: 10.3390/s24175807.